

Underwater Vision-Based Surveillance Using Machine Learning & AI

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REVISED RESEARCH QUESTION

How does underwater image enhancement affect the performance of deep-learning-based object detection across the three supplied underwater datasets?

RESEARCH FOCUS

- Compare original images vs. enhanced images
- Measure enhancement's effect on object-detection performance
- Use a transfer-learning-based YOLO detector as candidate approach
- Evaluate all three supplied datasets — separately, per class structure

Measurable Outputs & Target Definition

OBJECT DETECTION METRICS

- Precision
- Recall
- mAP@0.5
- mAP@0.5:0.95

IMAGE ENHANCEMENT METRICS (WHERE APPLICABLE)

- PSNR
- SSIM
- UIQM

COMPUTATIONAL PERFORMANCE

- Inference time per image / frame
- Enhancement processing time, if applicable

Main Comparison: *Original* → Detector vs. *Enhanced* → Detector

TARGET DEFINITION

Primary

Detect underwater objects — class, bounding box, confidence score — across all three datasets.

Secondary

Determine whether enhancement improves, has little effect, or degrades detection performance.

DATASET SCOPE — ALL THREE INCLUDED

well.v8i.yolov8	4 classes
DIATAquarium.v4i.yolov8	11 classes
Aquatic Plant.v2i.yolov8	1 class

Evaluated separately — class structures differ across datasets.

Completed Work & Key Findings

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DAY 1 — PROBLEM UNDERSTANDING

- Understood the underwater surveillance problem
- Inspected all three supplied datasets
- Identified classes, splits & YOLO annotations
- Defined pipeline: Input → Enhancement → Detection → Evaluation → Tracking (if applicable)

✓ Problem Understanding

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DAY 2 — LITERATURE REVIEW

Reviewed 7 closely related papers across enhancement, detection, deep learning, datasets & tracking.

Candidate methods shortlisted:

- White Balance / CLAHE
- Water-Net / FUnIE-GAN
- YOLO-based transfer-learning detector
- IoU / SORT-style tracker (if sequential data)

✓ Candidate Methods Identified

✓ Research Question Refined

KEY FINDINGS

- Underwater images suffer color distortion, low contrast, blur & poor visibility
- Project compares original vs. enhanced rather than assuming improvement
- Three datasets have different class structures — evaluated separately

- Literature shows enhancement does not always improve detection accuracy
- YOLO-based detection shortlisted as candidate approach
- Tracking remains conditional on availability of video/sequential data

Blockers, Mentor Confirmation & Next Plan

POINTS REQUIRING MENTOR CONFIRMATION

Dataset Configuration

Confirm the validation-path issue in Well & DIAT Aquarium data.yaml files (pointing to test/images instead of valid/images)

Evaluation Metrics

Confirm which detection and enhancement metrics are mandatory

Enhancement Comparison

Confirm whether the main experiment compares Original → Detection vs. Enhanced → Detection

Tracking

Confirm whether video/sequential data is available and whether tracking is required

NEXT PLAN

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Day 4 — Dataset Readiness

- Define schema, labels, targets & metadata
- Identify missing-data, imbalance, leakage risks
- Define correct train/valid/test split
- Prepare dataset-risk checklist

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Day 5 — Pipeline & Experiment Planning

- Design reproducible workflow
- Define baseline, advanced & project-specific experiment
- Finalize tentative evaluation criteria
- Create experiment log & repo structure

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Day 6 — Methodology & Mentor Review 2

- Complete literature, methodology & data-readiness package
- Present preprocessing, modelling & evaluation plan
- Confirm split strategy, metrics & constraints
- Prepare Methodology V1 & updated plan